Prototype of Early Fire Detection System for Home Monitoring Based On Wireless Sensor Network

Ferry Astika Saputra, M. Udin Harun Al Rasyid, Bey Aryo Abiantoro
Politeknik Elektronika Negeri Surabaya (PENS), Indonesia
ferryas, udinharun{@pens.ac.id}, beyaryo@gmail.com

***Abstract*— Fire is one of the few disasters that frequently occurred in Indonesia. Jakarta has archived 1139 fire incident with more than 20 casualties in 2016. By the rapid development of information technology and communication in Indonesia, it turns out that the number of incident and casualty still cannot be suppressed. This research is focused on early fire detection and home monitoring based on Wireless Sensor Network using 4 sensors such as temperature, air humidity, Carbon Monoxide, and smoke sensors. If the fire probability calculation categorized as high, system will perform an action of unlocking home door lock, turning on alarm, and notifying the user. Based on the trials and testing result, system is able to send notification to user precisely. However, error still can be encountered with up to 0.3% ratio on fire probability calculation.**

Keywords—Fire: E-Home, Wireless Sensor Network, Internet of Thing, Fuzzy Logic

I. INTRODUCTION

Fire occurrence does not know place and time, and can be happened anywhere and anytime. Almost every day through media's, we can witness fire in any place across country. Eventually, such disaster can also target our home.

Fire can damage property, stagger economy and business, and even taking casualties. Jakarta itself has archived 1139 fire incident in 2016 with more than 20 casualties [1]. The cause of such incident is varying, but mostly because of short circuit in electricity or human error. Fire maybe unexpected, but to take action to minimize the risk of fire occurrence or for the worst case that the fire cannot be avoided is absolutely required, furthermore we simply want to suppress the fire impact to the smallest point, hence the casualties can also be minimized.

When the fire taking place in house, the most important thing that we should put a concern on is to evacuate victims as many as possible, allowing them to escape safely. Thus, emergency exit access is playing the biggest role in this case. However, evacuation may find adversary if the emergency path is locked or unexpectedly difficult to reach.

We conduct a work to create home condition monitoring system which work automatically and able to detect any incident potential at the earliest state in order to prevent the disaster. Yet, fire is unpredictable and may gone wild at any time in that case system will unlock the door, providing emergency entrance and exit access so the evacuation will run smoothly. It is expected that the developed system will provide a solution for home condition monitoring and execute a necessary disaster response action in the effort of preventing any fallen casualties.

A lot of research regarding fire monitoring has been conducted recently. Chen et al. [2] proposed a system which detects fire in cargo compartment in an aircraft. Combination of smoke increment concentration, Carbon Monoxide (CO), and Carbon Dioxide (CO2) is captured and included in algorithm to increase the reliability of the system. Zhang et al. [3] also wrote about their research result. Their article taking up an explanation about smoke sensor usage and temperature sensor installed in building to monitor and detect fire point in earliest state.

Several sensor data possess complex attributes, ambiguous, and uncertain. Fuzzy set theory can be a solution in data modeling with the mentioned attribute. By combining Wireless Sensor Network (WSN) and fuzzy logic, the obtained data is more accurate and comprehensive in calculation [4-5, 8].

In general view, a sensor node is dependent on the battery in order to works. Because the power of the battery is limited, then the power usage should be noticed in most extreme way. Fahmi et al. [6] discussed about sleep schedule. Sleep schedule is useful to reduce power usage in node sensor by stopping all process in a specific time range according to obtained fuzzy value.

In order to make system more responsive, then interaction between user and system is required. Sen et al. [7] described a detail about how the systems which monitor fire transmit information to user through Short Message Service (SMS).

Sensor node is a small electronic device which has a limited storage. This is become problem for an application that required a continuous, long term, and secure for data acquisition. In order to solve that issue, the data must send to data center server. Huda et al. [9] proposed a system to transfer the data using ciphertext based encryption (CP-ABE). The implementation of CP-ABE provided secure data sensor sharing without causing overload on the system.

WSN is not just about monitoring environment condition. Some researches implement this technology in order to monitor wider area. Kumar et al. [10] proposed a research to monitor wildfire in the forest using temperature and humidity sensors. By using GPRS technology, the exact location of wildfire can be detected faster and more accurate. Al Rasyid et al. [11] is also uses WSN technology to determine water volume for agriculture. The result obtained from the study is proposed system was able to monitor and give the information about the watering volume so the plant area can get ideal humidity level.

Some previous works has made a reference for this research. We will focus on building a system which is more

effective than before. The contribution of this paper is to provide the environment condition and give an alarm when the condition is dangerous. System will be built in such way and allow user to monitor their house condition in real-time. Another unique feature from this research is the present of early step that will perform a necessary action when fire incident is at its edge. If the probability calculation can be categorized as high, system will active an emergency action by unlocking door, make the alarm go on, and notify the user in effort of avoiding any casualties. Besides that, the usage of power in sensor node is very important in order the node to be active continuously. Viewed from node sensor datasheet, the most used power is generated by data transmission process. Therefore, to reduce power usage of node sensor in addition of sleep scheduling, data will be included in data transmission algorithm and concluded whether data should be sent or not.

II. Proposed System

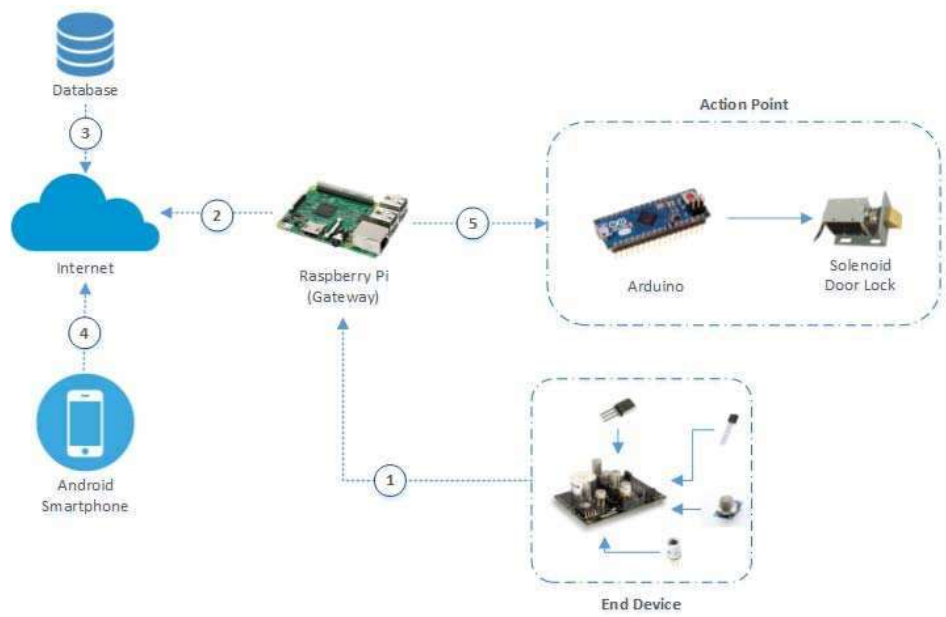


Fig. 1. Proposed system design

In this research, system is designed to focus on environment monitoring and taking early action to prevent any casualties. System consists of 5 primary points, which are end device, gateway, action point, server, and user. The generic design of the system can be found in Figure 1.

End device is built within 4 sensors (temperature, humidity, CO, and smoke sensors) that installed on gases sensor board of waspmote. End device will monitor temperature, humidity, CO, and smoke in indoor area. Obtained data is utilized to calculate fire probability value using rule-based fuzzy that has been made before. Room condition data and fuzzy value will be transmitted to gateway via XBee communication. The greater fire probability, the more intensive end device would monitor room condition. As Vice versa, the lesser fire probability, the more reduced monitoring activity rate. The prototype of end device node is shown in Figure 2.

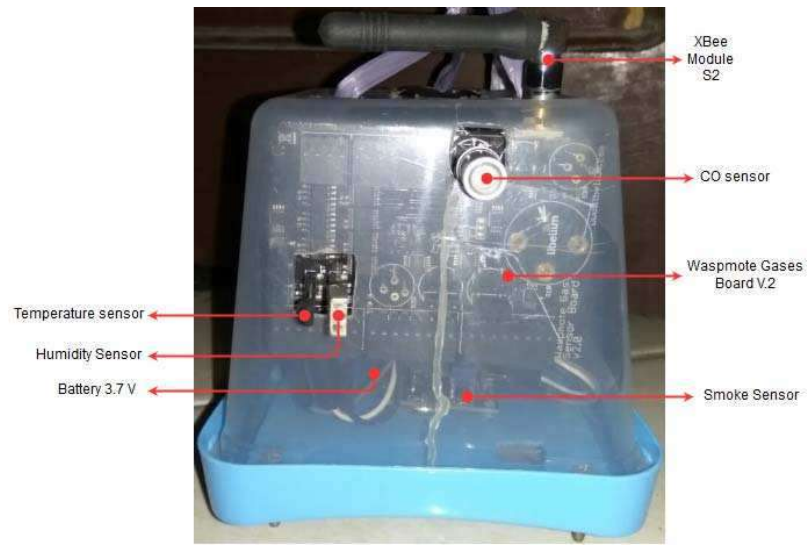


Fig. 2 Prototype of end device node

Fuzzy logic act importantly in combining data from the four sensors and generating fire probability output. The application of fuzzy logic aims to increase the reliability and accuracy from the system so it would reduce false alarm ratio that would be sent to user. Below is the design of the algorithm, and can be observed from Figure 3.

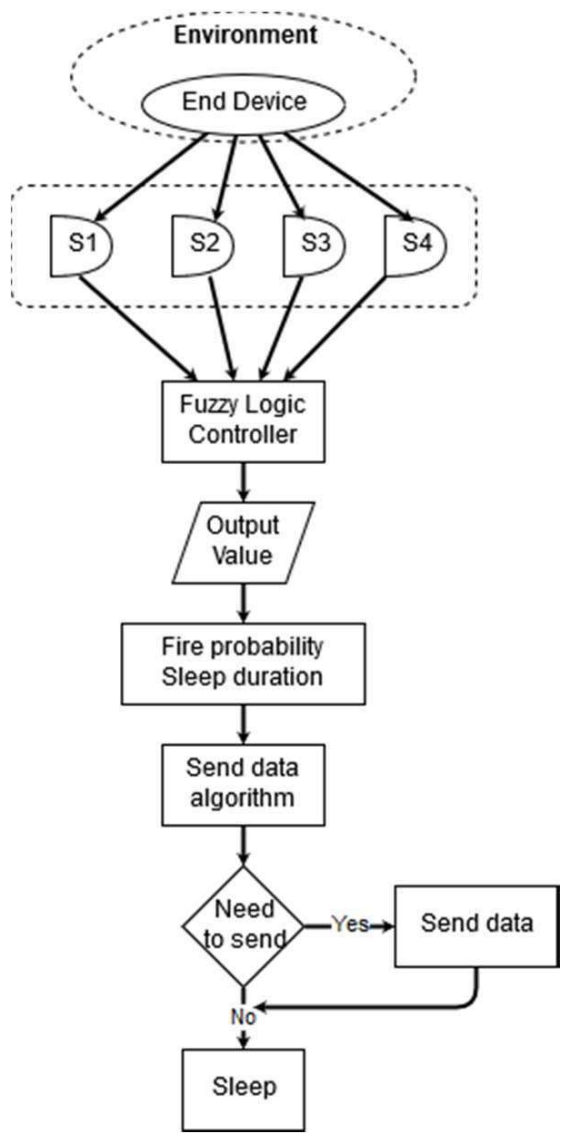


Fig. 3. Design algorithm

Details:

S1 = Temperature sensor (°C)
S2 = Humidity sensor (%RH)
S3 = Carbon monoxide sensor (ppm)
S4 = Smoke sensor (ppm)

Under our developed fire detection algorithm, temperature, humidity, smoke, and CO are the input of fuzzy variables and fire probability is the output variable. Membership function of LOW, MEDIUM, HIGH are set on each input variable and NORMAL, ALERT, FIRE are all set on output variable. Figure 4 to 7 show graphic membership on each input variable temperature, humidity, Carbon Monoxide, and smoke. Figure 8 shows graphic membership in fire probability variable output.

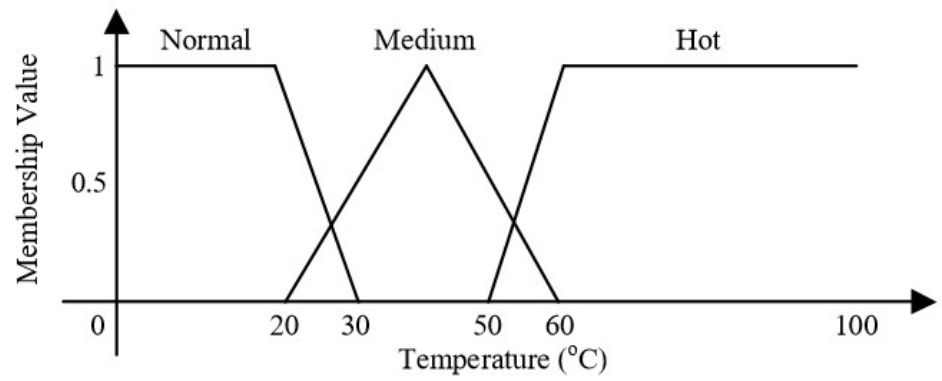


Fig. 4. Membership function for temperature

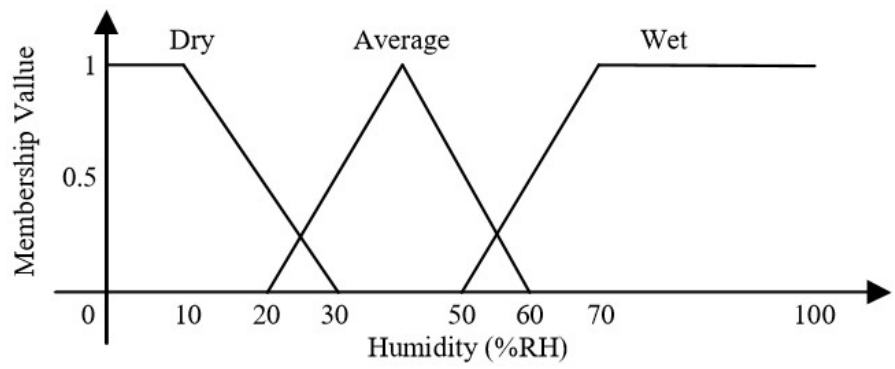


Fig. 5. Membership function for humidity

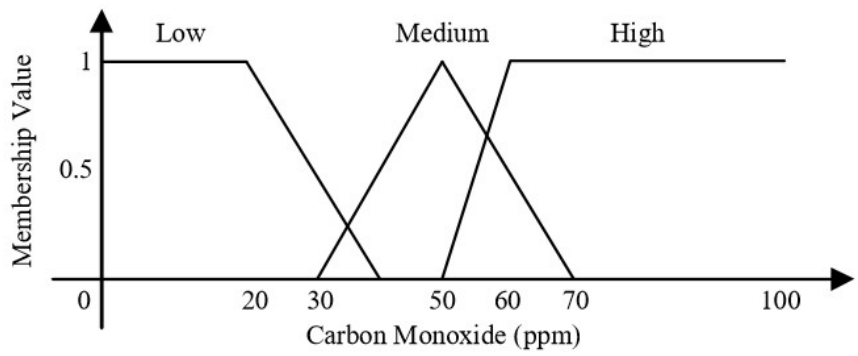


Fig. 6. Membership function for CO

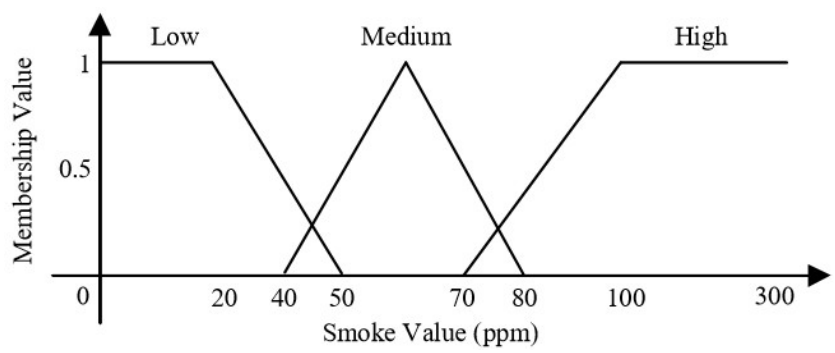


Fig. 7. Membership function for smoke

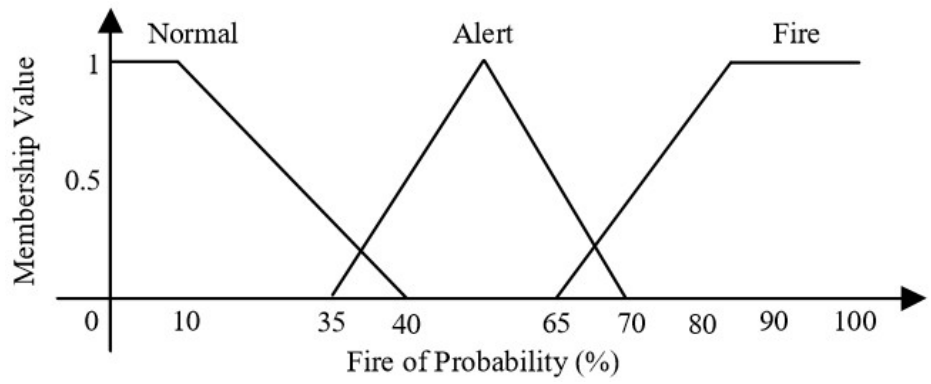


Fig. 8. Membership function for fire probability

Intervention system of fuzzy consists of collection of rules (IF condition THEN action). Gaussian, triangle, and trapezoid are most used function for membership function. The core of fuzzy system is existing on rule base system that resembles human reasoning. If-then rule in expert system is commonly described as below:

```
F x1 is A11 and x2 is A21 . . . THEN y is B1
else IF x1 is A12 and x2 is A22 . . . THEN y is B1
else . . .
IF x1 is A1n and x2 is A2n . . . THEN y is Bn
```

Where x_1, x_2, … are input variable, y is output variable, and A_{11} … A_{1n} are fuzzy set.

This research uses 4 input variable and each variable has 3 membership. So there are 3x3x3x3 = 81 used rule in total that obtained from every input possibility that may exist. Rule in this research can be observed from Table 1.

TABLE 1. Fuzzy logic rule base

No	Temperature	Humidity	CO	Smoke	Probability
1	normal	wet	low	low	normal
2	normal	wet	low	med	normal
3	normal	wet	low	high	alert
…	…	…	…	…	…
80	high	dry	high	med	fire
81	high	dry	high	high	fire

End device, being part of the system, has 4 sensors that continuously monitor environment. Data from the sensors is used to get fire probability value and sleep duration for end device. Sleep mode method is a method that used to increase power usage efficiency. When in sleep condition, device will be deactivated. The duration of sleep time from end device depends on output value that produced by fuzzy. The formula of sleep scheduling is described as below:

$$sTime = sMax-((vFuzzy/(vMxFuzzy - vMnFuzzy))*(sMax - sMin)) \quad (1)$$

Description:

sTime = Duration Sleep Time
sMax = Maximum Duration Sleep Time (10s)
sMin = Minimum Duration Sleep Time (0s)

vFuzzy = Fuzzy Output Value
vMxFuzzy = Maximum Fuzzy Output Value
vMnFuzzy = Minimum Fuzzy Output Value

If the data modeled as linear curve, then the formula above would appear like figure 9.

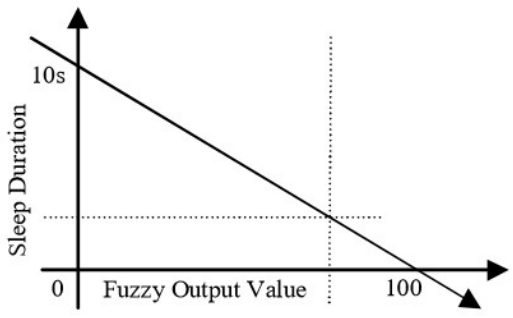


Fig. 9. Linear curve of sleep duration

Before being sent, the obtained data is collected into send data algorithm. From the algorithm itself that the data would be decided whether it should be sent out or not so the device usage of XBee communication would be more effective than before. Send data algorithm can be observed form Figure 10.

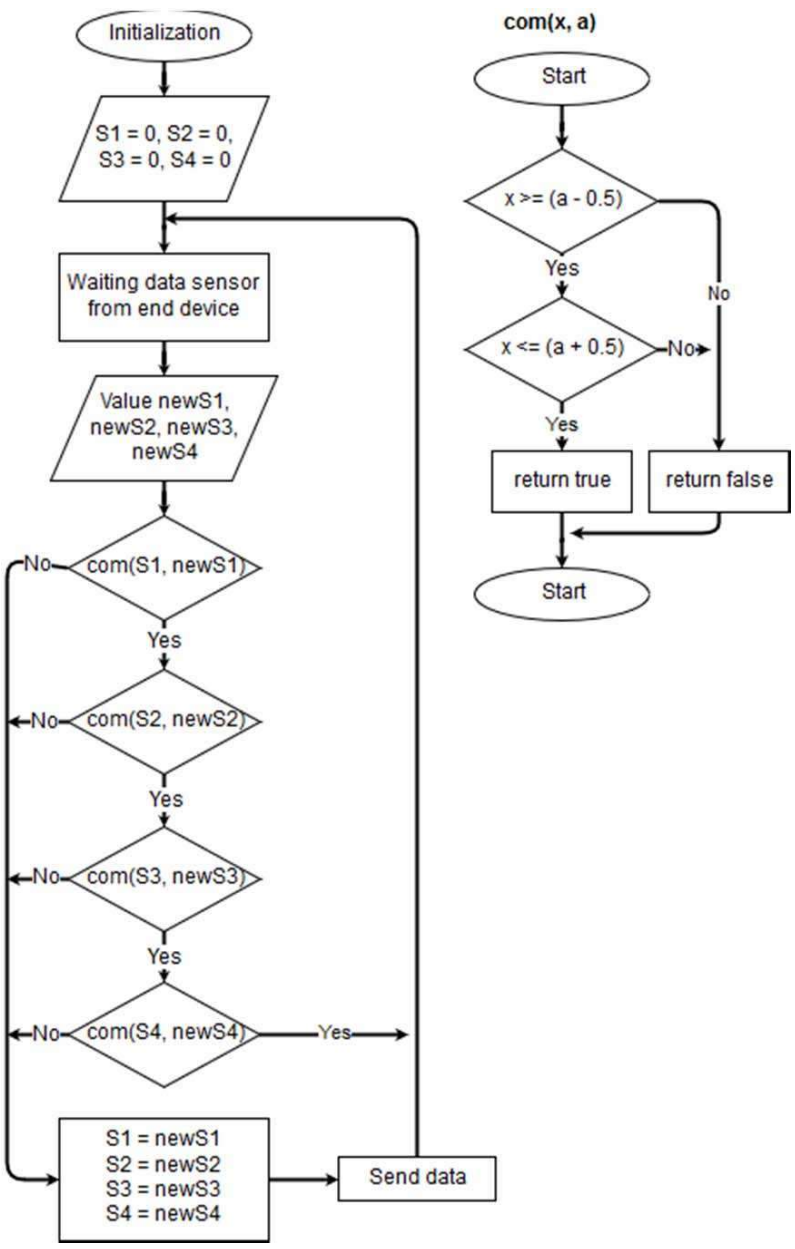


Fig. 10. Send data algorithm diagram

Gateway is composed from raspberry pi 3 board, XBee communication module, and buzzer that act as alarm. The Programming language that used in gateway is JavaScript. Sensor data that obtained from end device through XBee communication will be sending to server and stored into server database. Below is a photo of the appearance of gateway.

Fig. 11. Prototype of gateway node

If the fire probability value form obtained data is categorized as ALERT, gateway would switch on the alarm to alert people nearby within certain duration. And if the fire probability value is categorized as FIRE, gateway would continuously ring the alarm and transmit command broadcast to action point to unlock the door through XBee communication line.

Action point is composed from arduino board, XBee module, solenoid door lock, and a few more electronics components. The circuit scheme of the action point is shown by Figure 12.

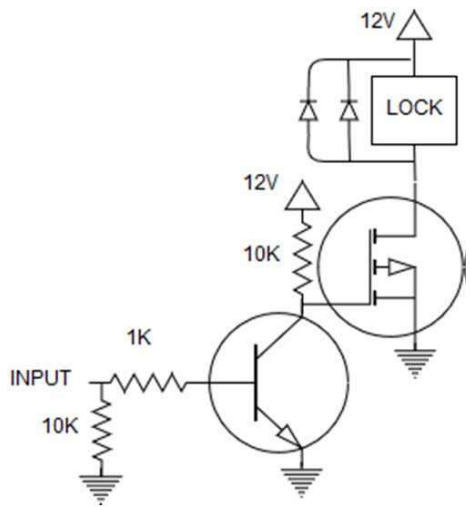


Fig. 12. Action point circuit scheme

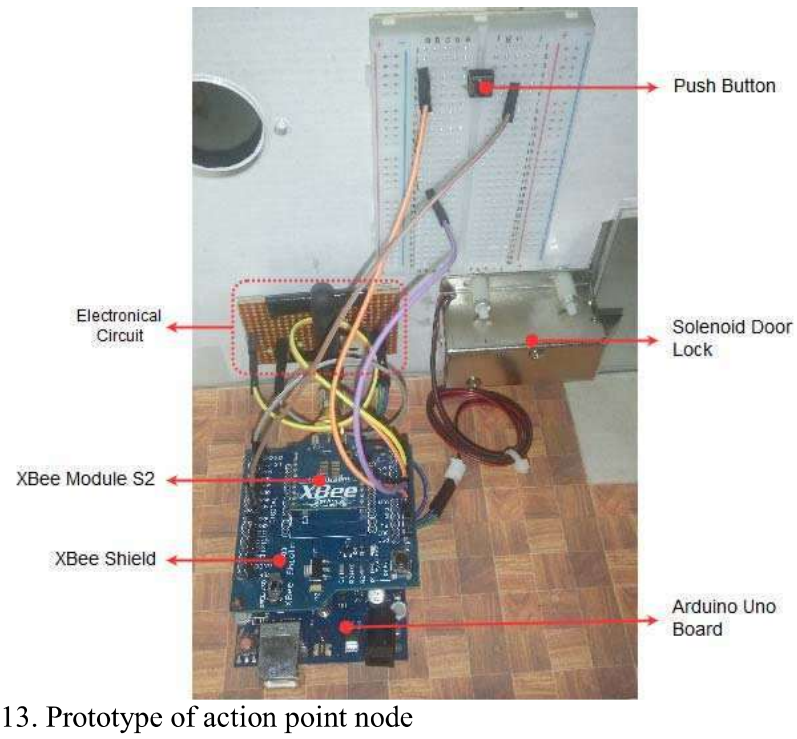


Fig. 13. Prototype of action point node

The initial state of the action point is always locked. There are 3 ways to change action point state become unlocked. The first one is through user command from android application to unlock the door. At this state the action point would be opened for 5 seconds and after that would be back to its original state. The second method is when user pressed the button in action point. Unlock button in action point is functioned in such way so the user is able to unlock the lock system when no internet connection is present (hybrid). The duration of the unlocking of the action point is equal to the duration of the first condition. The third condition is when the fire probability enter FIRE category. At this state, action point would be unlocked in continuous manner until the fire probability gradually dropped.

Server is built using JavaScript programming language. Server has several functions, store the data to the cloud database, access media for user, and as communication link between user and gateway.

In order to make user able to access data easier, the system also equipped with Android-based application as the main media. This application has capability to communicate with gateway in 2 ways, the first is through server and the second is direct communication to gateway if the user is located in the same local network. When the fire probability is increasing and finally reached the ALERT or FIRE category, server sends notifications to alert the user by using cloud messaging to the application.

III. Experimental Result

Fuzzy logic membership test is done first. Value of fuzzy input variable obtained from four sensors which are generated. Testing is done by simulating environment condition using JavaScript based program. Sensor values are obtained randomly by giving minimum and maximum limit for random value. Range of minimum and maximum value for four sensors are 0 to 100 (for temperature, humidity, and CO) and 0 to 300 (for smoke).

Test is done automatically by taking 30 sample data's. The result from this simulation is shown in table 2 below.

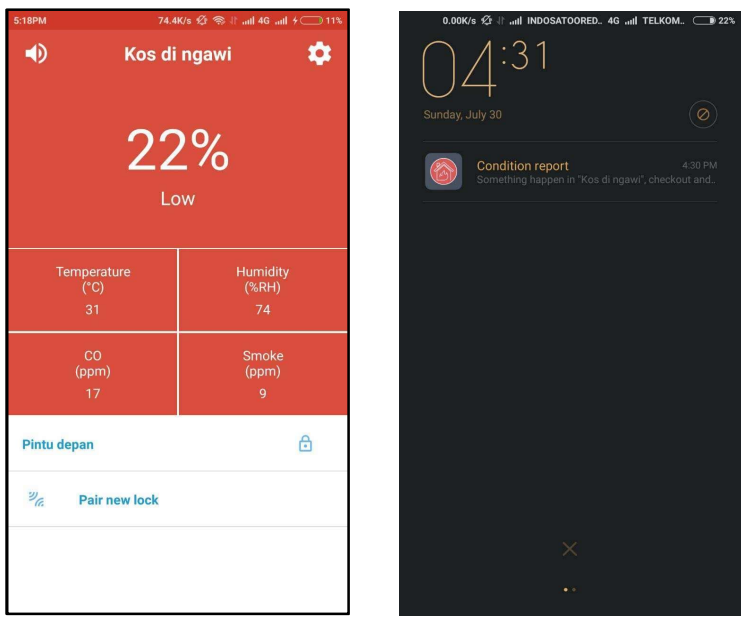
TABLE 2. Result of fuzzy logic membership simulation

no	temp	hum	co	smoke	fuzzy	category	desired category
1	19	91	38	239	57	alert	Alert
2	12	0	62	12	40	alert	Alert
3	47	78	9	24	21	normal	normal
4	24	71	18	9	27	normal	normal
5	11	38	31	204	57	alert	alert
6	82	86	35	8	48	alert	alert
7	21	0	6	189	44	alert	alert
8	90	96	41	151	85	fire	Fire
9	48	87	68	38	53	alert	alert
10	40	20	62	3	69	fire	Fire
11	5	6	85	230	100	fire	Fire
12	77	13	54	291	87	fire	Fire
13	61	6	78	136	100	fire	Fire
14	54	87	4	236	75	fire	Fire
15	6	84	46	33	23	normal	normal
16	29	3	4	271	81	fire	Fire
17	59	25	33	34	63	alert	alert
18	38	7	21	16	12	normal	normal
19	56	67	45	27	44	alert	alert
20	49	81	43	26	23	normal	alert
21	90	55	27	58	60	alert	alert
22	76	74	28	15	23	normal	normal
23	25	67	34	202	63	alert	alert
24	67	28	73	94	77	fire	fire
25	13	98	68	160	98	fire	fire
26	4	28	26	195	56	alert	alert
27	3	31	42	36	26	normal	normal
28	92	77	65	8	92	fire	fire
29	26	94	77	23	40	alert	normal
30	1	54	3	69	32	normal	normal

After testing, the experimental results were analyzed. The table above shows that there are 2 data categorized incorrectly which are data number 20 and 29 so we got error ratio equal to 6,67 % within calculation below :

$$\begin{aligned} \textit{Error value} &= (\textit{incorrect data} \,/\, \textit{total data}) \\ \textit{Error ratio} &= \textit{Error value} * 100\ \% \\ \textit{Error value} &= (2 / 30) \\ &= 0{,}06667 \\ \textit{Error ratio} &= 0{,}06667 * 100\ \% \\ &= 6{,}67\ \% \end{aligned}$$

In the client side, the application for Smartphone based on Android has been made. This application can display data both in real-time or log history and can warn the user by showing notifications when probability of fire in their environment has changed. Figure 14 shows some of the user interfaces of the application.



(a) (b)

Fig. 14. (a) Main user interface to show realtime data and (b) Notification from the application

IV. Conclusion

From the test that has been done, the proposed system has been able to send alarm to alert user precisely to prevent fire occurrence early with a 6.67 % error ratio. Android-based application used to show the data both in real time or log history also simplify user to monitor their environment condition. However, there are some obstacles in *send algorithm*. If monitored environment conditions change very quick, then the *send algorithm* is less effective to use. Moreover, fuzzy logic rule's is also needs to configured in further research to minimalized false alarm send to user.

References

[1] Statistik Kebakaran Berdasarkan Penyebab. (2017, July 25). Retrieved from *http://www.jakartafire.net/statistic*

[2] S.J. Chen, C. Hovde, and K.A Peterson, "Fire detection using smoke and gas sensors," Fire Safety Journal, 2007.

[3] L. Zhang, and G. Wang, "Design and Implementation of Automatic Fire Alarm System based on Wireless Sensor Networks," International Symposium on Information Processing, 2009.

[4] P. Manjunatha, A.K. Verma, and A. Srividya, "Multi-Sensor Data Fusion in Cluster based Wireless Sensor Networks Using Fuzzy Logic Method," IEEE Region 10 Colloqium and the Third ICIIS, 2008.

[5] V. Khanna, and R.K. Cheema, "Fire Detection Mechanism using Fuzzy Logic," International Journal of Computer Science, 2013.

[6] N. Fahmi, M.U.H. Al Rasyid, and A. Sudarsono, "Adaptive Sleep Scheduling for Health Monitoring System Based on the IEEE 802.14.3 Standard," EMITTER International Journal of Engineering Technology, 2016.

[7] K. Sen, J. Sarkar, S. Saha, A. Roy, D. Dey, S. Baitalik, and C.S. Nandi, "Automated Fire Detection And Controlling System," International Advanced Research Journal in Science, Engineering and Technology, 2015.

[8] K. Kumar, N. Sen, S. Azid, and U. Mehta, "A Fuzzy Decision in Smart Fire and Home Security System," IEEE International Symposium on Robotics and Intelligent Sensors, 2016.

[9] S. Huda, N. Fahmi, A. Sudarsono, and M.U.H Al Rasyid, "Secure Data Sensor Sharing on Ubiquitous Environmental Health Monitoring Application," Jurnal Teknologi (Sciences & Engineering), 2015.

[10] D.K Kumar, G. Kishore, and T.V.S Kumar, "Fire Monitoring System for Fire Detection Using Zigbee and GPRS System," IOSR Journal of Electronic and Communication Engineering, 2017.

[11] M.U.H Al Rasyid, E.M. Kusumaningtyas, and F. Setiawan, "Application to Determine Water Volume for Agriculture Based on Temperature & Humidity Using Wireless Sensor Network," Knowledge Creation and Inteligent Computing (KCIC), 2016.